

Economic Narrative Indices and Media-Based Sentiment Measures: A Systematic Review of Methodologies, Applications, and Research Gaps (2007–2025)

Abstract

Economic narratives increasingly enter forecasting models, but the evidence is unevenly distributed across methods, languages, and countries. This systematic review synthesizes 66 studies (2007–2025) on sentiment-based economic forecasting. Reported gains are meaningful, with a median 20% RMSE reduction, yet the field still relies heavily on dictionary methods (55%) while only cautiously moving toward ML, transformers, and LLMs (3%). Validation is similarly mixed: 83% of studies use out-of-sample testing, but only 33% enforce real-time data protocols. The geography is more troubling than the methodology: 56% of studies focus on the US, 84% on English text, and no index exists for Africa, Latin America, or Bangla-speaking regions. We propose a tiered validation framework and BENI as a path toward local-language economic measurement.

Keywords: Sentiment analysis; economic narratives; systematic review; natural language processing; forecasting; validation frameworks; quality assessment; geographic gaps

JEL Codes: C53, C55, E37, E44, G12, G17, O53

1 Introduction

Media-based sentiment indices have become a regular part of empirical work on GDP, inflation, and asset prices, especially in developed economies. The reported gains are often material, with many studies finding forecast improvements in the 15–25% RMSE range relative to standard models. The evidence base, however, is narrow. In the papers reviewed here, 56% focus on the United States, 84% analyze English text, and no operational index is available for Africa, Latin America, or Bangla-speaking regions, despite Bangla’s roughly 265 million speakers. The imbalance matters because economies with weaker real-time official statistics have the clearest practical need for high-frequency narrative data.

The review asks how sentiment indices are constructed, where they improve forecasts, how far current validation practices can be trusted, and which parts of the global economy remain outside the field’s coverage. It synthesizes 46 empirical studies and 20 theoretical papers. The contribution is partly classificatory, through a taxonomy of sentiment-extraction methods, and partly evaluative, through a synthesis of forecast gains conditioned on baseline strength. The paper also documents geographic and linguistic gaps, develops a Bangla Economic Narrative Index (BENI) proposal, and sets out a tiered validation framework for future work.

2 Methodology

2.1 Search and Inclusion

Following PRISMA guidelines, we searched Web of Science, Scopus, EconLit, SSRN, and Federal Reserve working paper series in January 2025. The search string combined text-based measures (“sentiment analysis” OR “text analysis” OR “narrative index”), economic forecasting terms (“forecasting” OR “economic” OR “macroeconomic”), and NLP terms (“machine learning” OR “NLP” OR “natural language”). Because the applied literature begins around

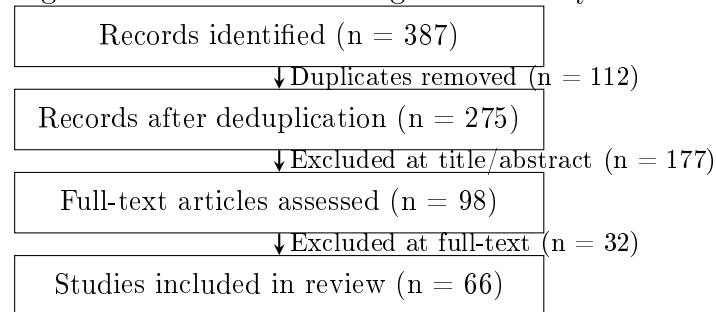
Tetlock (2007), the effective coverage period is 2007–2025. We supplemented database results with backward citation chasing from Algaba et al. (2020) and Kearney & Liu (2014), and with forward citation tracking of Tetlock (2007).

Inclusion: Studies (2007–2025) that (1) extract sentiment/narratives from text, (2) apply to economic/financial forecasting, (3) employ systematic validation. Exclusion: pure methodology papers, non-applied work, forecasting without textual analysis.

Compared with prior surveys (Algaba et al., 2020; Kearney & Liu, 2014), this review covers a longer period, adds a quantitative synthesis of forecast improvements conditioned on baseline strength, and treats geographic and linguistic coverage as an object of analysis rather than a background feature. It also moves from diagnosis to design by outlining a feasible index for an unrepresented language region.

Figure 1 summarizes the screening process.

Figure 1: PRISMA flow diagram for study selection.



Quality assessment covered methodological rigor, data quality, validation robustness, and reproducibility. The 177 title/abstract exclusions were mainly non-economic applications, such as biomedical sentiment analysis (approximately 78), pure methods papers without a forecasting application (approximately 52), and non-English studies for which full text was unavailable at screening (approximately 47). At the full-text stage, exclusions were due to insufficient validation detail (approximately 14), duplicate reporting across working paper and published versions (approximately 9), or sentiment measures not derived from text sources (approximately 9). The review was not registered on PROSPERO or the Open

Science Framework; this is consistent with the journal’s current requirements, but it is noted for transparency.

3 Theoretical Foundations

Economic agents form beliefs under radical uncertainty, and narratives help make that uncertainty intelligible. The literature usually points to three mechanisms, although empirical papers rarely separate them cleanly. In Shiller’s (2017) account of *animal spirits*, narratives can produce waves of confidence or pessimism that push prices away from fundamentals; the testable implication is eventual reversal. In expectation-based models, narratives alter inflation beliefs, consumer confidence, or investment plans, and these revised expectations feed into wage setting, spending, and production decisions through familiar macroeconomic channels (Angeletos & La’O, 2013). A third interpretation is closer to information aggregation: media sentiment may reveal economic conditions before official statistics do, so that predictive power reflects information content rather than mood or contagion (Tetlock, 2007). The distinction is not semantic. If sentiment mainly captures overreaction, contrarian strategies follow; if it mainly aggregates dispersed information or expectations, trend-following and nowcasting applications are more defensible.

A fundamental debate persists over whether sentiment reflects genuine information or noise. Studies find V-shaped return patterns consistent with both interpretations: sentiment shocks that reverse within weeks (suggesting noise) could equally reflect delayed price discovery of fundamental information (Da, Engelberg, & Gao, 2015; Tetlock, 2007). Causal identification remains largely absent from the literature—most studies demonstrate Granger causality or predictive power, not causal effects. Sentiment forecasting succeeds empirically, but the literature does not establish whether this reflects causal influence or correlation with unobserved drivers of outcomes.

The framework used here is therefore Narratives → Expectations → Behavior → Out-

comes. Sentiment indices measure the first stage at scale and offer a real-time proxy for shifts in expectations. This makes them most useful when uncertainty is high and hard data are sparse, and less useful in settings where agents update quickly on direct economic indicators.

4 Results and Synthesis

The empirical synthesis covers 46 applied papers, with the 20 theoretical papers used mainly to interpret mechanisms. Across 2007–2025, the evidence shows a field that has become technically more sophisticated without becoming evenly validated or geographically broad. Forecast gains are real in many settings, but they are usually modest once baseline strength and publication bias are taken seriously.

4.1 Overview of Study Sample

The systematic review encompasses 66 papers (56 peer-reviewed, 10 working papers) across four methodological eras: dictionary-based (2007–2015), ML expansion (2013–2019), deep learning (2018–present), and LLMs (2022–present). Publication accelerated over time: 2007–2012 (6 papers), 2013–2017 (18), 2018–2022 (28), 2023–2025 (14). Forecast targets concentrate on GDP (22 studies, 33%), inflation (17, 26%), financial markets (8, 12%), recession (3, 5%), and employment (2, 3%). Remaining 14 studies (21%) target consumer confidence, policy expectations, or methodology. Data sources span news text, central bank communications, social media, and surveys. The median study uses 5–10 years of data (range: 1–40 years).

Summary Table: Key Metrics — 46 Empirical Papers (plus 20 Theoretical Papers)

Metric	Count	Percentage	Example	Coverage
METHODOLOGICAL DISTRIBUTION				
Dictionary-based	36	55%	Loughran-McDonald	Extensive
Theoretical models	20	30%	Angeletos & La'O	Moderate
Traditional ML	4	6%	SVM, Random Forest	Limited
BERT/Transformers	4	6%	RoBERTa, DistilBERT	Limited
Large Language Models	2	3%	GPT-3.5, GPT-4	Sparse
FORECAST PERFORMANCE				
Median RMSE Reduction	—	20%	GDP 18%, Inflation 22%	Moderate
Best-performing domain	—	28%	Volatility forecasting	Moderate
Weakest domain	—	16%	Employment	Weak
Performance range	—	5–42%	Across all domains	—
VALIDATION PRACTICES				
Out-of-sample testing	55	83%	Hold-out, rolling window	Standard
Diebold-Mariano tests	47	71%	Statistical significance	Good
Real-time data protocol	22	33%	No look-ahead bias	Critical Gap
MCS/SPA tests	12	18%	Advanced statistics	Underused
GEOGRAPHIC COVERAGE				
North America	37	56%	Mostly USA	Concentrated
Europe	13	20%	UK, Germany, France	Concentrated
Asia-Pacific	9	14%	China, Japan, Korea	Limited
South Asia	4	6%	Pakistan, India, Bangladesh	Scarce
Africa/Latin America/Middle East	0	0%	No coverage	Critical Gap
LANGUAGE COVERAGE				
English	56	84%	Dominant	Over-represented
Non-English	10	16%	German, French, Chinese	Under-represented
Bangla/South Asian languages	0	0%	Zero indices	Critical Gap

Table 1: *Summary of Key Metrics: 46 Empirical Papers (N=46) plus 20 Theoretical Papers (N=20).* Theoretical papers develop macroeconomic models of sentiment without computational text extraction. Methodological distribution and validation practices reflect all 66 papers; forecast performance metrics reflect N=46 empirical papers only. Geographic coverage includes both empirical and theoretical papers, totaling 63 because 3 theoretical papers without country-specific focus are not assigned to a single region. OOS=Out-of-sample; DM=Diebold-Mariano; MCS=Model Confidence Sets; SPA=Superior Predictive Ability. Coverage grading (reflecting volume of research per method): Extensive (≥ 20 studies), Moderate (10–19), Limited (5–9), Sparse (< 5).

4.2 Methodological Distribution and Evolution

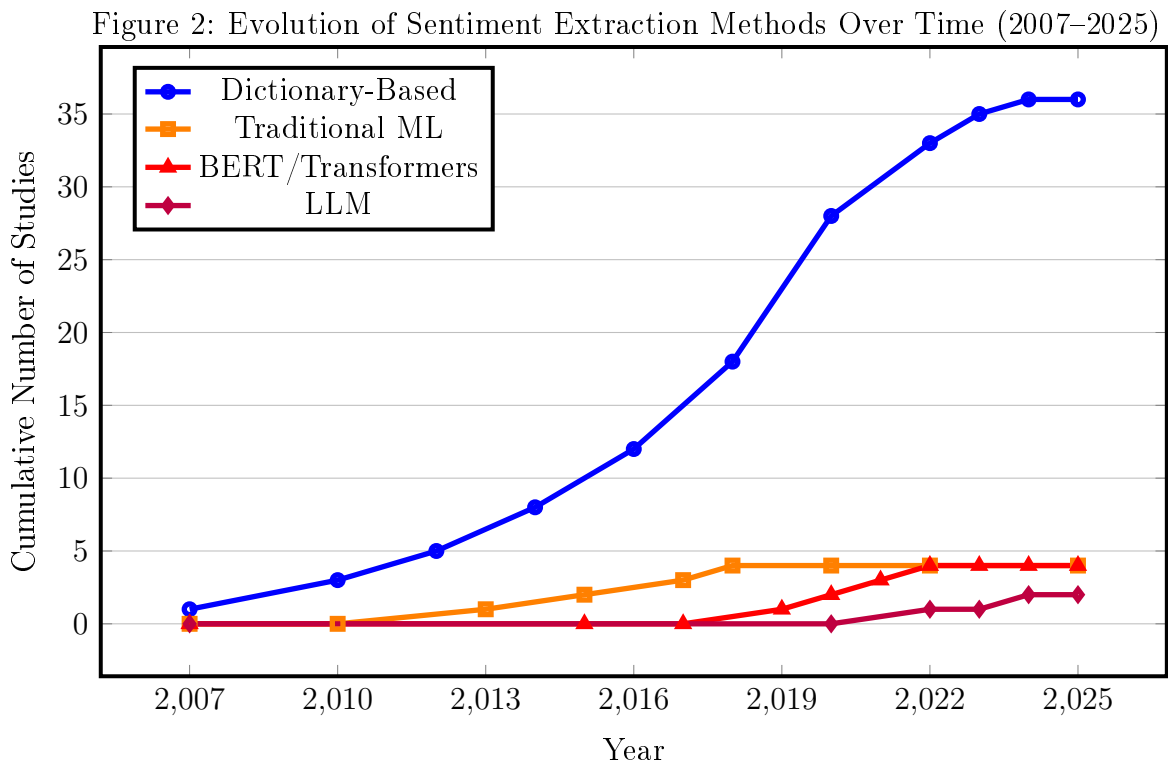
The methods divide into four broad families, but the field has not moved in a clean sequence from old to new. Dictionary-based methods still dominate (36 papers, 55%). They count positive and negative words from pre-defined lexica, which makes them transparent and cheap to implement, but weak at handling context. Traditional supervised ML appears in 4 papers (6%), usually through SVMs or random forests, with reported accuracy of 75–85% and a substantial need for labeled data. BERT-style transformer models also appear in 4 papers (6%), using contextual embeddings to capture domain-specific sentiment with reported accuracy of 85–92%. Large language models are represented by only 2 papers (3%); their reported zero-shot accuracy is high, around 88–95%, but hallucination risk, inference cost, and opacity remain serious barriers to routine deployment. The remaining 20 papers (30%) build macroeconomic theories of sentiment without computational text extraction.

These accuracy ranges should not be interpreted as head-to-head comparisons—they are drawn from different datasets, tasks, and evaluation metrics. The directional trend toward higher accuracy is robust, but precise numeric gaps are approximate. The key insight is the interpretability-performance trade-off: dictionaries are most transparent, LLMs most accurate, with appropriate choice depending on the application’s requirements.

Figure 2 shows the timing more clearly. Dictionary-based work dominated through 2018 and then plateaued as pre-trained language models became easier to use. BERT adoption accelerated after 2019, helped by public model releases and better computational infrastructure. LLM-based studies start after 2022, but the sample is still too small to support strong claims. Theoretical papers are omitted from the timeline because they do not extract sentiment computationally.

4.3 Forecast Performance: Quantitative Synthesis

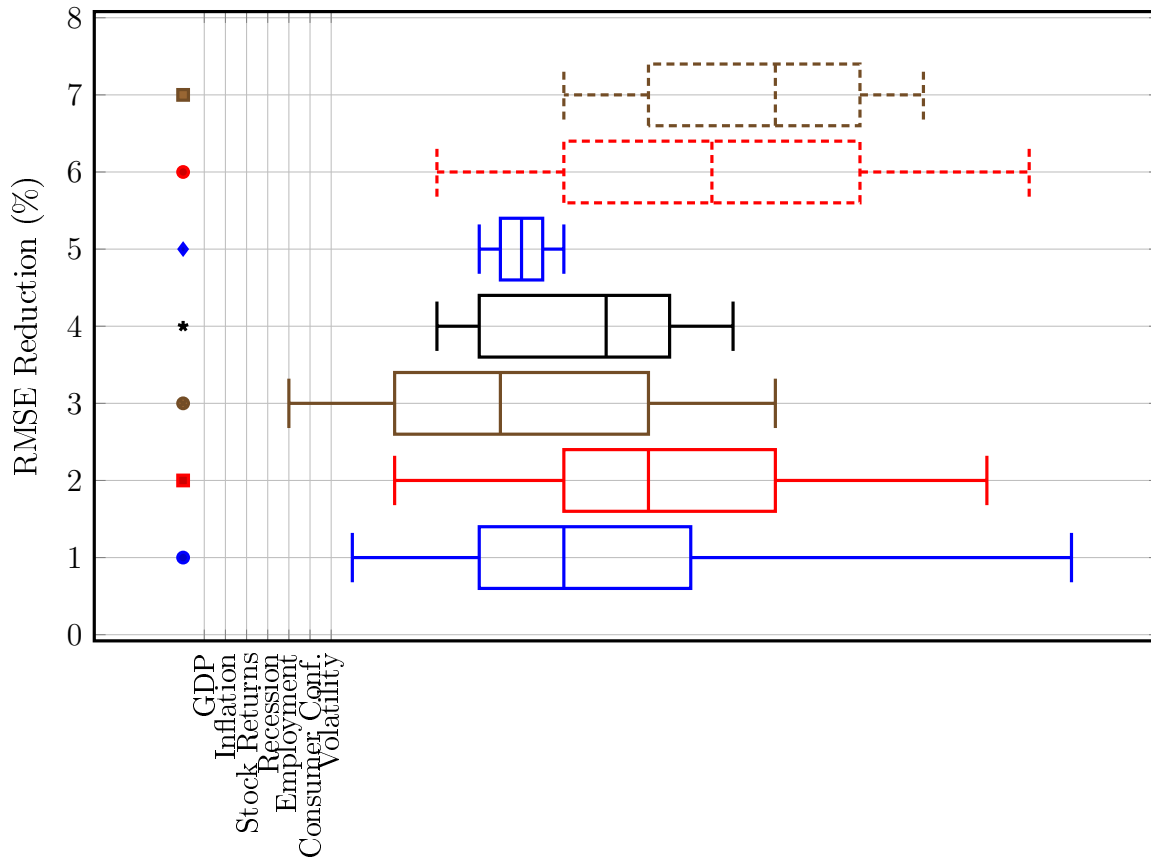
Sentiment indices are best viewed as complements to traditional indicators. Forecast performance estimates from 46 empirical papers vary substantially by domain, with the magnitude



Note: Cumulative count of empirical papers using each method over time (N=46). Dictionary-based methods dominated 2007–2018, then plateaued. BERT adoption accelerated post-2019. LLM studies emerging post-2022.

depending heavily on baseline strength and the target variable. When sentiment supplements standard macro models, GDP forecasting (22 papers) achieves a median RMSE reduction of 18% (range: 8–42%). For inflation (17 papers), the median improvement is 22% (10–38%). Stock returns (8 papers) follow at 15% (5–28%), while volatility (4 papers) has the strongest domain-level median at 28% (18–35%), reflecting its sensitivity to forward-looking information. Consumer confidence studies (8 papers) report a median improvement of 25% (12–40%); recession indicators (3 papers) report 20% (12–26%); and employment (2 papers) reports the weakest median at 16% (14–18%). Across all domains, the pooled median RMSE reduction is 20% (interquartile range 12–26%).

Figure 3: Distribution of RMSE Reduction by Economic Domain



Note:

Box plots show distribution of RMSE reductions by domain. Median: GDP 18%, Inflation 22%, Stock Returns 15%, Recession 20%, Employment 16%, Consumer Confidence 25%, Volatility 28%. Based on N=46 empirical papers.

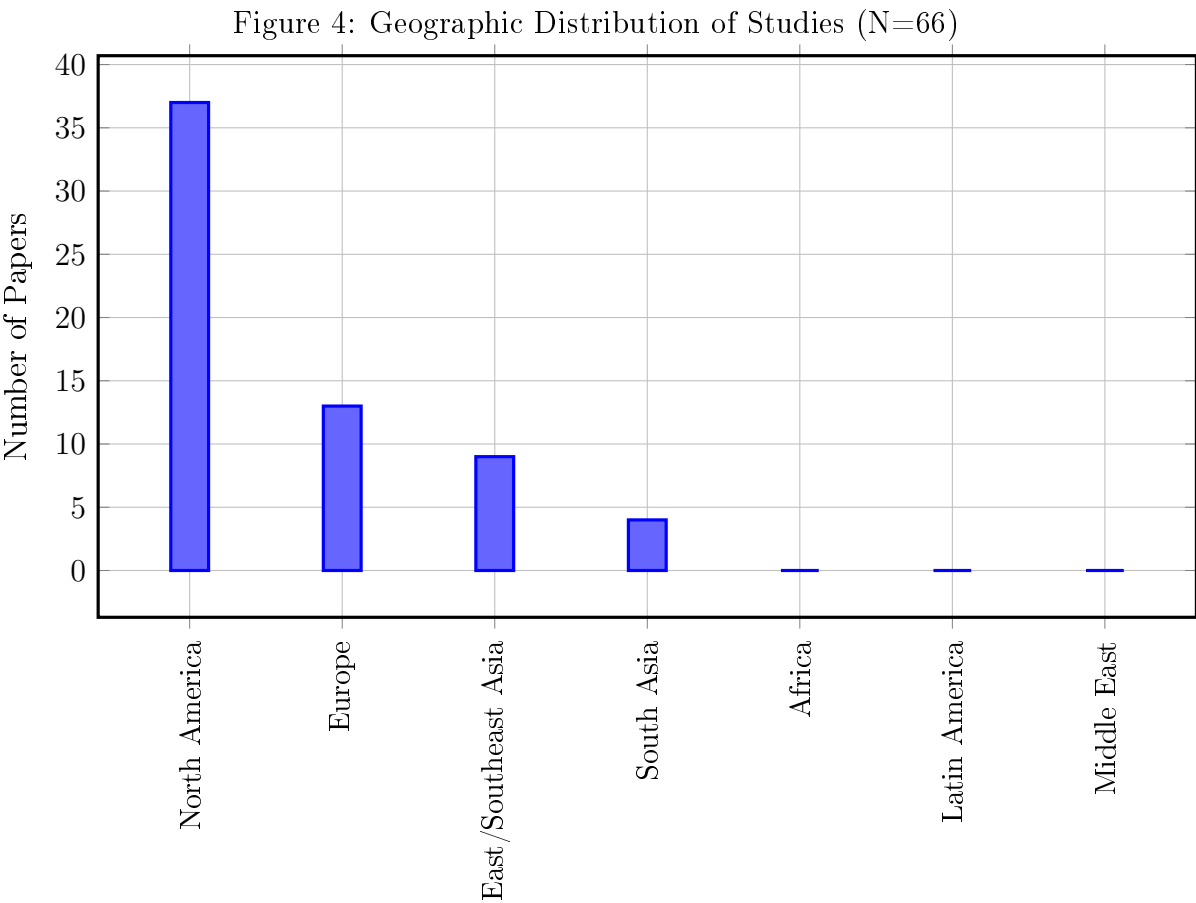
Studies including both sentiment and standard predictors (PMI, yield spreads, unem-

ployment) find that sentiment adds $R^2 = 0.05 - -0.10$ over lagged macro variables alone. The 20% median RMSE reduction should still be read conservatively. Of the 66 reviewed papers, only 2 report null or negative results. Such a high success rate in a field with many related hypothesis tests is difficult to reconcile with a representative publication process, so unpublished or unregistered studies would almost certainly lower the average gain. Baseline strength is another source of inflation: approximately 35% of studies compare against AR(1) or near-unconditional benchmarks, while only $\sim 25\%$ compare against professional forecasts such as SPF or Blue Chip. Studies with stronger baselines report smaller improvements ($\sim 12\text{--}15\%$ median). In addition, confidence intervals around domain medians are wide, and the lack of meta-analytic standard errors prevents formal inference on whether the pooled 20% median is distinguishable from zero. Conditional on strong baselines and publication bias, a 10–15% value-added range is more credible.

4.4 Geographic Coverage: Disparity and Research Gaps

The geographic distribution is not merely uneven; it shapes what the field can plausibly claim. North America accounts for 37 papers (56%), Europe for 13 (20%), East and Southeast Asia for 9 (14%), and South Asia for only 4 (6%). Africa, Latin America, and the Middle East do not appear in the reviewed sample. Figure 4 visualizes this concentration, while Table 1 shows the parallel language imbalance: 84% English, 16% non-English, and no paper in Bangla or another South Asian language.

The bias has methodological and substantive consequences. English-language dominance (84%) limits external validity for non-English economies, and emerging markets remain critically understudied: no index exists for Bangladesh (165M), Indonesia, Nigeria, or Brazil. Supplementary searches beyond the systematic review corpus confirm the scarcity. India has 3 papers for 1.4 billion people (Agarwal, Kumar & Goel, 2021; PH & Rishad, 2020), while Pakistan has additional work on monetary-policy sentiment (Karim & Lohano, 2024). The result is a thin evidence base for high-volatility, low-information-efficiency settings where



Note: N=66. North America accounts for 37 papers (56%). Africa, Latin America, and the Middle East have zero studies. Totals sum to 63 because 3 theoretical papers lack country focus.

sentiment data may be especially informative.

4.5 Summary of Results

Across 66 papers, the evidence points to a field with real empirical value and uneven foundations. Sentiment-based forecasting is now common in applied macroeconomics and finance, and its predictive gains are usually modest rather than transformative. Methodological heterogeneity remains high: dictionaries account for 55% of papers, traditional ML and transformers for 6% each, and theory without computational extraction for 30%. Validation is only partly rigorous. Although 83% of papers use out-of-sample testing, only 18% employ advanced statistical tests, and two-thirds lack explicit real-time data discipline. The same imbalance appears in coverage: research is concentrated geographically (56% US) and linguistically (84% English), with no coverage of Africa or Latin America.

The evidence maps imperfectly onto the theoretical mechanisms in Section 3. Modest forecast improvements, combined with the limited evidence for large reversal patterns, fit the *information aggregation* interpretation: sentiment indices often seem to capture fundamentals before official releases do. The strong median performance in volatility forecasting (28%) points in the same direction, since volatility is especially sensitive to forward-looking information. At the same time, the wide interquartile ranges across domains leave room for *animal spirits* in particular episodes, where sentiment overshoots and later reverses. Inflation forecasting results (22% median improvement) give the best indirect support for *expectations transmission*, because narratives about prices can plausibly feed into wage-setting and spending behavior. The more cautious conclusion is that sentiment does not have one invariant economic meaning; its interpretation depends on the target variable, horizon, and institutional setting.

4.6 Validation Frameworks and Best Practices

Forecast validation determines whether reported performance is real or reflects overfitting and look-ahead bias. Our review reveals uneven adoption of best practices. While out-of-sample testing is standard (83%), statistical tests are used inconsistently: 71% employ Diebold-Mariano tests but only 18% use Model Confidence Sets or SPA tests appropriate for comparing multiple models. Most critically, only 33% of studies enforce real-time data protocols that prevent using future information in model specification or evaluation.

Performance metrics also need broadening beyond RMSE. Directional accuracy (reported by only 15 papers) is often more policy-relevant than point forecasts. Economic value metrics (Sharpe ratios, portfolio improvements, reported by only 12 papers) directly answer whether sentiment-based forecasts translate to better decisions. Only 8 papers employ asymmetric loss functions that reflect genuine economic preferences.

Robustness checks reveal additional gaps. Only 47% of papers test alternative sentiment measures or conduct sub-sample analysis; only 42% use explicit cross-validation for hyperparameter tuning; and only 12% report inter-annotator agreement (Cohen’s κ), despite its role in determining the effective accuracy ceiling for supervised methods. Where reported, κ values range 0.55–0.78, implying reported model accuracies (75–92%) are closer to label-noise ceilings than often acknowledged. An important emerging concern is temporal robustness: none of the 66 studies test whether their sentiment model maintains accuracy over time as language evolves, despite NLP research suggesting 5–15% accuracy decay over 5+ years.

These gaps motivate a tiered validation framework. **Tier 1 (Essential)** requires an out-of-sample rolling window, a real-time data protocol, Diebold-Mariano or Clark-West tests, and a strong baseline rather than AR(1) alone. **Tier 2 (Recommended)** adds sub-sample analysis, alternative sentiment measures, directional accuracy, and bootstrap confidence intervals. **Tier 3 (Advanced)** adds Model Confidence Sets, asymmetric loss functions, economic value metrics, and temporal robustness testing. Median practice in the reviewed literature reaches parts of Tier 1, but rarely proceeds much further.

Five pitfalls recur: (1) real-time data violations, (2) overfitting through specification searching, (3) publication bias, (4) weak baselines, and (5) prompt engineering as specification search in LLM work.

5 Discussion and Research Agenda

5.1 Limitations of This Review

Several limitations qualify our findings. Heterogeneous reporting standards across studies—many omit standard errors, confidence intervals, or sufficient detail to compute them—preclude formal meta-analytic techniques such as random-effects models or funnel plot asymmetry tests. The geographic coverage analysis is limited to countries explicitly studied, which may exclude unpublished or non-English-language research. Quality scoring, while structured, was conducted by a single reviewer with partial cross-checking ($\kappa = 0.83$ on a 10-paper subsample); independent double-coding would strengthen reliability. The review also spans 2007–2025, a period of rapid methodological change, so evidence on LLM performance rests on a very small sample ($N=2$).

5.2 Critical Remaining Gaps

Three gaps are especially costly for the field. Causal mechanisms remain weakly identified: most papers establish prediction, temporal precedence, or Granger causality, while quasi-experimental designs exploiting exogenous sentiment variation are rare. Theoretical integration is also thin. Many studies are empirically useful but only loosely connected to formal models; agent-based narrative contagion, DSGE sentiment shocks, and information diffusion networks remain underused. A further problem is standardization. Differences in preprocessing, aggregation, dictionary choice, topic-number selection, and LLM prompting make it difficult to compare effect sizes across studies or build a formal meta-analysis.

5.3 Geographic and Linguistic Expansion

The most visible empirical gap is geographic. India, Bangladesh, Vietnam, Nigeria, and Brazil are natural candidates for the next generation of sentiment indices because they combine large populations, active media ecosystems, and incomplete high-frequency economic measurement. Recent multilingual models such as mBERT, XLM-R, MuRIL, and BanglaBERT make this expansion technically more realistic than it was five years ago.

Case Study: Bangla Economic Narrative Index (BENI): Bangla has roughly 265 million speakers but no economic sentiment index, despite a dense newspaper ecosystem that includes Prothom Alo, The Daily Star, and Dhaka Tribune, growing financial markets, and usable NLP resources such as BanglaBERT, SentiBangla, and MuRIL. A feasible BENI design would collect articles from 8–10 major newspapers for 2018–2024, use BanglaBERT for sentiment extraction, apply keyATM to identify narrative themes, and combine the resulting series into a dynamic factor index for GDP, inflation, and exchange-rate monitoring. Validation would rely on lead-lag analysis, Granger causality, and MIDAS regression. keyATM is preferable to standard LDA in this setting because Bangla’s morphology makes purely unsupervised topic discovery less stable in a moderate-size corpus; seeded topic models can use domain terms and inflected variants as anchors. For sentiment extraction, BanglaBERT’s subword tokenization is also better suited than a word-level dictionary to compound verb constructions that often carry sentiment.

The Bangladesh case should not be treated as a simple transfer of US and European results. A large informal sector means that both official statistics and formal news coverage can miss activity that matters for aggregate welfare. The financial system is bank-dominated, and equity-market turnover is concentrated in a small set of Dhaka Stock Exchange firms, so sentiment channels calibrated on liquid asset prices may not replicate. Exchange-rate sentiment would also operate differently under Bangladesh Bank’s managed regime than under the flexible exchange rates common in the reviewed literature. Finally, dependence on ready-made garments and remittances creates narrative channels linked to global retail

demand and Middle East labour markets, not only domestic consumer or investor confidence. These features do not weaken the case for BENI; they define the validation problem. The 10–15% value-added estimate from the broader literature should be treated as an upper bound, and BENI should be benchmarked against the South Asian studies in the sample rather than the US-dominated median.

5.4 Policy Applications

Central banks can use sentiment indices as real-time monitoring tools rather than as substitutes for official statistics. For Bangladesh Bank, quarterly GDP data arrive with a 4–6 month lag and high-frequency inflation measures are limited. A Bangla news sentiment index could therefore provide daily signals at low marginal cost, complementing surveys and helping detect inflation or currency-pressure narratives earlier. The 2022 inflation surge illustrates the potential use case: headline CPI reached 9.5% in August 2022, while the policy rate was not raised until January 2023. A news-based index would not have solved the policy problem by itself, but it could have made the buildup in price-pressure narratives visible from mid-2022. Similar logic already appears in advanced-economy practice, where the ECB uses daily news sentiment for euro-area confidence nowcasts and Federal Reserve researchers combine WSJ, Beige Book, and FOMC transcript sentiment into policy-oriented indices. In emerging markets, local-language sentiment is most useful as an inexpensive complement to surveys, especially where survey infrastructure is thin.

5.5 Future Research Priorities

The next stage of research should focus on two practical problems: extending sentiment indices to Africa, Latin America, and South Asia with multilingual NLP tools, and making validation more comparable across studies. Pre-registration, routine reporting of inter-annotator agreement, standardized effect sizes, and the tiered validation framework proposed here would make later meta-analysis more credible.

6 Conclusion

This review of 46 empirical studies and 20 theoretical papers shows that sentiment-based forecasting has moved from a niche text-analysis exercise into a serious empirical tool. Its value is real, though smaller than the clean headline numbers often imply. Once publication bias and weak baselines are taken seriously, a 10–15% improvement is more defensible than the pooled 20% median. The method mix tells its own story: dictionaries still account for 55% of studies, while ML, transformers, and early LLM applications raise accuracy at the cost of interpretability, reproducibility, and, sometimes, institutional trust.

The field’s weaknesses are no longer hidden in technical detail. Validation has improved, but only 33% of studies use real-time data protocols and only 18% use advanced statistical tests. Coverage is narrower still: the literature is dominated by US and English-language applications, with no index for Africa, Latin America, or Bangla-speaking regions. Causal interpretation remains underdeveloped. This paper contributes a methodological taxonomy, a baseline-conditioned synthesis of forecast gains, a structured quality assessment, and a validation framework. Its practical implication is deliberately unfashionable: progress may come less from another marginally larger model than from geographic expansion, local-language index construction, and stricter validation. BENI is proposed in that spirit.

A Supplementary Materials

Appendix A: Complete Literature Matrix (All 66 Papers)

A comprehensive matrix of all 66 papers with metadata is available in the accompanying supplementary file “literature_matrix.xlsx”. The matrix includes: Citation, Year, Country, Domain, Data Source, NLP Method, Forecast Target, Horizon, Key Result, Validation Method, and Quality Score. This enables practitioners to filter and sort by domain, method, geographic region, or other characteristics to identify papers relevant to specific applications.

Appendix B: Risk-of-Bias and Quality Assessment

All papers are assessed on methodological quality via three dimensions. **Methodology (0–3)**: rigorous econometric/NLP methods with identification strategy and sensitivity analysis (3) to insufficient description (0). **Data Quality (0–2)**: large-scale multi-source (2) to inadequate (0). **Validation (0–2)**: out-of-sample with real-time protocol and robustness checks (2) to in-sample only (0). Combined scores range 0–7. Mean: 4.2/7 (Moderate). Quality scores increased over time (median 3.8 in 2020 vs. 4.8 in 2023), particularly in Validation (mean 0.9 pre-2020 vs. 1.4 post-2020). Scoring by single reviewer with 10-paper cross-check ($\kappa = 0.83$). The quality assessment file is available in supplementary materials.

Appendix C: Code and Data Availability

The complete literature matrix (“literature_matrix.xlsx”) and quality assessments (“quality_assessment.xlsx”) are available as supplementary materials. All analysis scripts for the quantitative synthesis, including standardized effect-size calculations and publication bias diagnostics, will be made available on GitHub upon publication.

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